

# TLUSI Supplementary Materials

TLUSI: Twin Learning Using Statistical Invariants with Distributional and Medical Predicates for Disease Diagnosis

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This supplement contains the complete proofs of Theorems 3.1 and 3.2 and Proposition 3.1, the complete disease-specific predicate selections, and the distributional predicate library. Statements, assumptions and equation references follow the corresponding manuscript. Tables A.1, B.1 and B.2 contain the predicate materials; Appendices C, D and E contain the proofs.

## Contents

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>A</b> | <b>Distributional Predicate Library</b>                   | <b>2</b> |
| <b>B</b> | <b>Selected Predicates for the Linear and RBF Kernels</b> | <b>3</b> |
| <b>C</b> | <b>Proof of Theorem 3.1</b>                               | <b>5</b> |
| <b>D</b> | <b>Proof of Proposition 3.1</b>                           | <b>5</b> |
| <b>E</b> | <b>Proof of Theorem 3.2</b>                               | <b>6</b> |

## Appendix A Distributional Predicate Library

Table A.1: Distributional predicate library and parameter-estimation procedures.

| Distribution | Probability density or mass function                                                                                                                                 | Parameters                                                         | Parameter estimation                                                                                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal       | $p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$                                                                                   | Location $\mu$ ; scale $\sigma > 0$                                | $\widehat{\mu} = \bar{x}$ , $\widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ .                                                                                        |
| Student- $t$ | $p(x) = \frac{\Gamma((\nu+1)/2)}{\sigma\sqrt{\nu\pi}\Gamma(\nu/2)} \left[1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right]^{-(\nu+1)/2}$                                      | Location $\mu$ ; scale $\sigma > 0$ ; degrees of freedom $\nu > 0$ | Set $\widehat{\mu} = \bar{x}$ and $\widehat{\sigma} = s\sqrt{(\nu-2)/\nu}$ , estimate $\nu \in [2.01, 200]$ by bounded maximum likelihood, where $s$ is the sample standard deviation. |
| Logistic     | $p(x) = \frac{\exp[-(x-\mu)/s]}{s[1 + \exp[-(x-\mu)/s]]^2}$                                                                                                          | Location $\mu$ ; scale $s > 0$                                     | $\widehat{\mu} = \bar{x}$ and $\widehat{s} = \widehat{\sigma}\sqrt{3}/\pi$ , using $\text{Var}(X) = \pi^2 s^2/3$ .                                                                     |
| Fisher $F$   | $p(x) = \frac{\Gamma((d_1+d_2)/2)}{\Gamma(d_1/2)\Gamma(d_2/2)} \left(\frac{d_1}{d_2}\right)^{d_1/2} \frac{x^{d_1/2-1}}{(1 + \frac{d_1}{d_2}x)^{(d_1+d_2)/2}}, x > 0$ | Degrees of freedom $d_1, d_2 > 0$                                  | Numerical maximum likelihood on positive-clipped samples, using a log-parameterization and bounds $0.05 \leq d_1, d_2 \leq 200$ .                                                      |
| Uniform      | $p(x) = \begin{cases} 1/(b-a), & a \leq x \leq b, \\ 0, & \text{otherwise} \end{cases}$                                                                              | Lower bound $a$ ; upper bound $b$                                  | $\widehat{a} = \min_i x_i$ and $\widehat{b} = \max_i x_i$ .                                                                                                                            |
| Beta         | $p(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, 0 < x < 1$                                                                                                                | Shape parameters $a, b > 0$                                        | Method of moments on $[0, 1]$ -clipped samples: $\widehat{a} = \bar{x}q$ , $\widehat{b} = (1-\bar{x})q$ , where $q = \bar{x}(1-\bar{x})/\widehat{\sigma}^2 - 1$ .                      |
| Gamma        | $p(x) = \frac{x^{a-1}\exp(-x/b)}{\Gamma(a)b^a}, x > 0$                                                                                                               | Shape $a > 0$ ; scale $b > 0$                                      | Numerical maximum likelihood initialized by $a_0 = \bar{x}^2/\widehat{\sigma}^2$ and $b_0 = \widehat{\sigma}^2/\bar{x}$ .                                                              |
| Weibull      | $p(x) = \frac{b}{a} \left(\frac{x}{a}\right)^{b-1} \exp\left[-\left(\frac{x}{a}\right)^b\right], x \geq 0$                                                           | Scale $a > 0$ ; shape $b > 0$                                      | Profile maximum likelihood: optimize $b \in [0.05, 50]$ and set $\widehat{a} = \left(n^{-1} \sum_{i=1}^n x_i^b\right)^{1/\widehat{b}}$ .                                               |
| Laplace      | $p(x) = \frac{1}{2b} \exp\left(-\frac{ x-\mu }{b}\right)$                                                                                                            | Location $\mu$ ; scale $b > 0$                                     | $\widehat{\mu} = \text{median}(x_1, \dots, x_n)$ and $\widehat{b} = n^{-1} \sum_{i=1}^n  x_i - \widehat{\mu} $ .                                                                       |
| Pareto       | $p(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, x \geq x_m$                                                                                                          | Lower bound $x_m > 0$ ; shape $\alpha > 0$                         | For positive-clipped samples, $\widehat{x}_m = \min_i x_i$ and $\widehat{\alpha} = \frac{n}{\sum_{i=1}^n \log(x_i/\widehat{x}_m)}$ .                                                   |
| GEV          | $p(x) = \frac{1}{\sigma} t(x)^{-1/\xi-1} \exp[-t(x)^{-1/\xi}], t(x) = 1 + \xi \frac{x-\mu}{\sigma} > 0$                                                              | Location $\mu$ ; scale $\sigma > 0$ ; shape $\xi$                  | Match the first three L-moments. When $\xi \rightarrow 0$ , use the Gumbel limiting density $\sigma^{-1} \exp[-z - \exp(-z)]$ , $z = (x-\mu)/\sigma$ .                                 |
| Triangular   | $p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)}, & a \leq x < c, \\ \frac{2(b-x)}{(b-a)(b-c)}, & c \leq x \leq b, \\ 0, & \text{otherwise} \end{cases}$              | Lower bound $a$ ; upper bound $b$ ; mode $c$                       | Set $\widehat{a} = \min_i x_i$ and $\widehat{b} = \max_i x_i$ ; estimate $\widehat{c}$ as the midpoint of the highest-frequency histogram bin.                                         |
| Bernoulli    | $p(x) = (1-p)\mathbb{1}\{x=a_0\} + p\mathbb{1}\{x=a_1\}$                                                                                                             | Support $\{a_0, a_1\}$ ; probability $p$                           | Class-conditional maximum likelihood: $\widehat{p} = n^{-1} \sum_{i=1}^n \mathbb{1}\{x_i = a_1\}$ .                                                                                    |
| Categorical  | $p(x) = \sum_{r=1}^R p_r \mathbb{1}\{x = a_r\}$                                                                                                                      | Support $\{a_r\}_{r=1}^R$ ; probabilities $\mathbf{p}$             | Class-conditional empirical frequencies: $\widehat{\mathbf{p}} = n^{-1} \sum_{i=1}^n \mathbb{1}\{x_i = a_r\}$ , with $\sum_{r=1}^R \widehat{p}_r = 1$ .                                |

## Appendix B Selected Predicates for the Linear and RBF Kernels

Tables B.1 and B.2 report the complete class-conditional predicates selected by TLUSI for each disease dataset. The notation and construction of the distributional and medical predicates follow the definitions in Section 4. In the tables,  $f_{\mathcal{D}}(v; \theta_c)$  denotes the class-conditional density or probability mass of distribution  $\mathcal{D}$  fitted in the current training fold,  $\widehat{F}_v$  denotes the empirical distribution function of  $v$ ,  $\sigma(t) = 1/(1 + \exp(-t))$ , and  $z_G = \text{PC1}(G)$  denotes the standardized first principal component of feature group  $G$ , and Med and Dis denote medical and distributional predicates, respectively.

Table B.1: Predicates selected by TLUSI using the linear kernel.

| Dataset                         |          | Predicate                     | Mathematical form                                                                                                             |
|---------------------------------|----------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Diabetes<br>768 × 8             | Positive | Med:InsBin                    | $\phi_+(x) = \phi_+^{\text{post}}(x_{\text{Ins}})$ , bins [0, 80, 150, 250, ∞)                                                |
|                                 |          | Med:PregStep                  | $\phi_+(x) = \mathbb{1}\{x_{\text{Preg}} \geq 5\}$                                                                            |
|                                 | Negative | Med:BMIBin                    | $\phi_+(x) = \phi_+^{\text{post}}(x_{\text{BMI}})$ , bins [0, 18.5, 25, 30, 35, 40, ∞)                                        |
|                                 |          | Med:INSBinMean                | $\phi_-(x) = \phi_-^{\text{con}}(x_{\text{Ins}})$                                                                             |
| Parkinson’s<br>195 × 22         | Positive | Dis:G1_GlucoseInsulin_PC1:GEV | $\phi_-(x) = f_{\text{GEV}}(z_{\text{GI}}; \widehat{\theta}_-)$                                                               |
|                                 | Negative | Med:BMISkinPCAMean            | $\phi_-(x) = \phi_-^{\text{con}}(z_{\text{BS}})$                                                                              |
| Thyroid<br>215 × 5              | Positive | Med:D2BinMean                 | $\phi_+(x) = \phi_+^{\text{con}}(x_{D2})$                                                                                     |
|                                 | Negative | None                          | ---                                                                                                                           |
| Heart failure<br>299 × 12       | Positive | Med:HypoPrimary               | $\phi_+(x) = \mathbb{1}\{x_{T4} \leq L_{T4}, x_{TSH} \geq H_{TSH}\}$                                                          |
|                                 | Negative | None                          | ---                                                                                                                           |
| Arrhythmia<br>452 × 15          | Positive | Med:CrBinMean                 | $\phi_+(x) = \phi_+^{\text{con}}(x_{\text{Cr}})$ , bins (−∞, 1.2, 1.5, 2.0, ∞)                                                |
|                                 | Negative | Med:CrPCAMean                 | $\phi_-(x) = \phi_-^{\text{con}}(z_{\text{Cr}})$                                                                              |
| Liver disorders<br>345 × 5      | Positive | Med:StableCardioRenal         | $\phi_-(x) = \mathbb{1}\{x_{\text{Cr}} < 1.2, x_{\text{EF}} > 40\}$                                                           |
|                                 | Negative | Med:WideQRS120                | $\phi_+(x) = \mathbb{1}\{x_{\text{QRS}} \geq 120\}$                                                                           |
| Liver patient<br>583 × 10       | Positive | Med:QRSBinMeanNeg             | $\phi_-(x) = \phi_-^{\text{con}}(x_{\text{QRS}})$                                                                             |
|                                 | Negative | None                          | ---                                                                                                                           |
| Breast cancer<br>286 × 9        | Positive | Dis:Feature_1:Logistic        | $\phi_-(x) = f_{\text{Logistic}}(z_{\text{MCV}}; \widehat{\theta}_-)$                                                         |
|                                 | Negative | Med:ASTHighRank               | $\phi_+(x) = \sigma(\widehat{F}_{\text{AST}}(x_{\text{AST}}) - 0.80)/0.05)$                                                   |
| Endometrial cancer<br>414 × 11  | Positive | Med:TBHighRank                | $\phi_+(x) = \sigma(\widehat{F}_{\text{TB}}(x_{\text{TB}}) - 0.80)/0.05)$                                                     |
|                                 | Negative | None                          | ---                                                                                                                           |
| Prostate cancer<br>2000 × 16    | Positive | Med:NodalCapsSoft             | $\phi_+(x) = \sigma((z_{\text{nodes}} + z_{\text{caps}} - \widehat{\theta})/\gamma)$                                          |
|                                 | Negative | Med:LowClinicBurden           | $\phi_-(x) = \mathbb{1}\{x_{\text{tumor}} < 20, x_{\text{nodes}} < 3, x_{\text{caps}} = \text{no}, x_{\text{grade}} \leq 2\}$ |
| Heart failure risk<br>3000 × 11 | Positive | Med:HighFGACNHigh             | $\phi_+(x) = \mathbb{1}\{x_{\text{FGA}} \geq 0.252475, s = 1\}$                                                               |
|                                 | Negative | Dis:Race_PC1:Categorical      | $\phi_+(x) = \mathbb{1}\{h \in \{2, 3\}, s = 1\}$                                                                             |
| Arrhythmia<br>452 × 15          | Positive | Med:NonEndoCNHigh             | $\phi_+(x) = f_{\text{Cat}}(z_{\text{Race}}; \widehat{\theta}_+)$                                                             |
|                                 | Negative | Med:LowFGACNLow               | $\phi_-(x) = \mathbb{1}\{x_{\text{FGA}} < 0.0499, s = 2\}$                                                                    |
| Heart failure<br>299 × 12       | Positive | Med:MutationSubtypePCAHigh    | $\phi_-(x) = \mathbb{1}\{\widehat{F}_{z_{\text{MS}}}(z_{\text{MS}}) \geq 0.75\}$                                              |
|                                 | Negative | Dis:PC1_G05:Bernoulli         | $\phi_+(x) = f_{\text{Bernoulli}}(z_{G05}; \widehat{\theta}_+)$                                                               |
| Arrhythmia<br>452 × 15          | Positive | Dis:PC1_G04:GEV               | $\phi_-(x) = f_{\text{GEV}}(z_{G04}; \widehat{\theta}_-)$                                                                     |
|                                 | Negative | Med:MorphRank                 | $\phi_-(x) = 1 - \frac{1}{4} \sum_{v \in \mathcal{M}} \widehat{F}_v(x_v)$                                                     |
| Arrhythmia<br>452 × 15          | Positive | Med:HighCreatinine            | $\phi_+(x) = \sigma((x_{\text{Cr}} - 1.5)/\gamma_{\text{Cr}})$                                                                |
|                                 | Negative | Dis:PC1_G11:Beta              | $\phi_-(x) = f_{\text{Beta}}(z_{G11}; \widehat{\theta}_-)$                                                                    |
| Arrhythmia<br>452 × 15          | Positive | Dis:PC1_G04:Bernoulli         | $\phi_-(x) = f_{\text{Bernoulli}}(z_{G04}; \widehat{\theta}_-)$                                                               |
|                                 | Negative | Dis:PC1_G09:GEV               | $\phi_-(x) = f_{\text{GEV}}(z_{G09}; \widehat{\theta}_-)$                                                                     |

*Note.* The subscripts GI, BS, Cr, and MS denote the Glucose–Insulin, BMI–SkinThickness, creatinine, and Mutation–Subtype representations, respectively. In the endometrial-cancer predicates,  $h$  and  $s$  denote the encoded histological type and molecular subtype. For MorphRank,  $\mathcal{M} = \{\text{radius, perimeter, compactness, symmetry}\}$ . All empirical quantities and distribution parameters are estimated exclusively from the corresponding training fold.

Table B.2: Predicates selected by TLUSI using the RBF kernel.

| Dataset                   |          | Predicate                      | Mathematical form                                                                                                                                   |
|---------------------------|----------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Diabetes<br>768 × 8       | Positive | None                           | ---                                                                                                                                                 |
|                           | Negative | Med:BMIBinMean                 | $\phi_-(x) = \phi_-^{\text{con}}(x_{\text{BMI}})$                                                                                                   |
| Parkinson’s<br>195 × 22   | Positive | Dis:G6_MDVP_Fhi_Hz_:GEV        | $\phi_+(x) = f_{\text{GEV}}(x_{\text{Fhi}}; \widehat{\theta}_+)$                                                                                    |
|                           | Negative | Med:PitchEntropy SpreadPCAMean | $\phi_-(x) = \phi_-^{\text{con}}(z_{\text{PES}})$                                                                                                   |
| Thyroid<br>215 × 5        | Positive | Med:HormonePC1Low              | $\phi_+(x) = \mathbb{1}\{z_{\text{H}} \leq t_{\text{H}}^{\text{low}}\}$                                                                             |
|                           | Negative | Med:TRHResponseCore            | $\phi_-(x) = \mathbb{1}\{L_{\text{TRH}} < x_{\text{TRH}} < H_{\text{TRH}}\}$                                                                        |
| Heart failure<br>299 × 12 | Positive | Med:AnaemiaRenal               | $\phi_+(x) = \mathbb{1}\{x_{\text{anaemia}} = 1\} \sigma((x_{\text{Cr}} - 1.5)/\gamma_{\text{Cr}})$                                                 |
|                           | Negative | Med:StableTriad                | $\phi_-(x) = \sigma((1.2 - x_{\text{Cr}})/\gamma_{\text{Cr}} + (x_{\text{EF}} - 40)/\gamma_{\text{EF}} + (x_{\text{Na}} - 135)/\gamma_{\text{Na}})$ |
| Arrhythmia<br>452 × 15    | Positive | Med:EFBinMean                  | $\phi_-(x) = \phi_-^{\text{con}}(x_{\text{EF}})$                                                                                                    |
|                           | Negative | Med:AgeBinMean                 | $\phi_-(x) = \phi_-^{\text{con}}(x_{\text{Age}})$                                                                                                   |
| Arrhythmia<br>452 × 15    | Positive | Med:CrPCAMean                  | $\phi_-(x) = \phi_-^{\text{con}}(z_{\text{Cr}})$                                                                                                    |
|                           | Negative | None                           | ---                                                                                                                                                 |
| Arrhythmia<br>452 × 15    | Positive | Dis:G1_AgeBodyMass_PC1:Weibull | $\phi_-(x) = f_{\text{Weibull}}(z_{\text{AB}}; \widehat{\theta}_-)$                                                                                 |
|                           | Negative | Dis:G1_AgeBodyMass_PC1:Weibull | $\phi_-(x) = f_{\text{Weibull}}(z_{\text{AB}}; \widehat{\theta}_-)$                                                                                 |

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Table B.2 continued

| Dataset                         |          | Predicate                                | Mathematical form                                                                                                             |
|---------------------------------|----------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Liver disorders<br>345 × 5      | Positive | Med:MCVBinMeanPos<br>Med:GGTBinMeanPos   | $\phi_+(x) = \phi_+^{\text{con}}(x_{\text{MCV}})$<br>$\phi_+(x) = \phi_+^{\text{con}}(x_{\text{GGT}})$                        |
|                                 | Negative | None                                     | --                                                                                                                            |
| Liver patient<br>583 × 10       | Positive | Med:HepaticInjury<br>PCAEvidencePos      | $\phi_+(x) = \phi_+^{\text{ncon}}(z_{\text{HI}})$                                                                             |
|                                 | Negative | None                                     | --                                                                                                                            |
| Breast cancer<br>286 × 9        | Positive | None                                     | --                                                                                                                            |
|                                 | Negative | Med:LowClinicBurden                      | $\phi_-(x) = \mathbb{1}\{x_{\text{tumor}} < 20, x_{\text{nodes}} < 3, x_{\text{caps}} = \text{no}, x_{\text{grade}} \leq 2\}$ |
| Endometrial cancer<br>414 × 11  | Positive | Med:HistologyPosteriorPos                | $\phi_+(x) = \phi_+^{\text{post}}(x_{\text{histology}})$                                                                      |
|                                 | Negative | Med:LowFGACNLow<br>Med:POLEUltraMutation | $\phi_-(x) = \mathbb{1}\{x_{\text{FGA}} < 0.0499, s = 2\}$<br>$\phi_-(x) = \mathbb{1}\{s = 4, x_{\text{mutation}} \geq 630\}$ |
| Prostate cancer<br>2000 × 16    | Positive | Med:MorphConcordant                      | $\phi_+(x) = \prod_{v \in \mathcal{M}} \mathbb{1}\{x_v \geq \tau_v\}$                                                         |
|                                 |          | Med:G03BinContrast                       | $\phi_+(x) = \phi_+^{\text{con}}(z_{G03})$                                                                                    |
|                                 |          | Med:PSAFreeConcordant                    | $\phi_+(x) = \mathbb{1}\{x_{\text{PSA}} \geq 10, x_{\text{freePSA}} < 0.15\}$                                                 |
|                                 | Negative | Med:PSARank                              | $\phi_-(x) = 1 - \widehat{F}_{\text{PSA}}(x_{\text{PSA}})$                                                                    |
|                                 |          | Dis:PC1_G03:GEV                          | $\phi_-(x) = f_{\text{GEV}}(z_{G03}; \widehat{\theta}_-)$                                                                     |
|                                 |          | Med:MorphRank                            | $\phi_-(x) = 1 - \frac{1}{4} \sum_{v \in \mathcal{M}} \widehat{F}_v(x_v)$                                                     |
| Heart failure risk<br>3000 × 11 | Positive | Med:AgeRank                              | $\phi_-(x) = 1 - \widehat{F}_{\text{Age}}(x_{\text{Age}})$                                                                    |
|                                 |          | Med:CardioRenalHigh                      | $\phi_+(x) = \sigma((40 - x_{\text{EF}})/\widehat{\gamma}_{\text{EF}} + (x_{\text{Cr}} - 1.5)/\widehat{\gamma}_{\text{Cr}})$  |
|                                 |          | Med:NaBinMean                            | $\phi_+(x) = \phi_+^{\text{con}}(x_{\text{Na}})$                                                                              |
|                                 |          | Med:AgeBinMean                           | $\phi_+(x) = \phi_+^{\text{con}}(x_{\text{Age}})$                                                                             |
|                                 | Negative | Med:LowCreatinine                        | $\phi_-(x) = \sigma((1.2 - x_{\text{Cr}})/\widehat{\gamma}_{\text{Cr}})$                                                      |

*Note.* Here,  $z_G = \text{PC1}(G)$ . PES, H, and HI denote the Pitch–Entropy–Spread, Hormone, and Hepatic-Injury representations, respectively; AB denotes the Age–BodyMass group,  $s$  is the encoded molecular subtype, and  $\mathcal{M} = \{\text{radius, perimeter, compactness, symmetry}\}$ . All empirical quantities and distribution parameters are estimated exclusively from the corresponding training fold.

## Appendix C Proof of Theorem 3.1

### Theorem 3.1 (statement).

Under the above conditions, the empirical invariants converge uniformly to their population counterparts. Specifically, for the positive class, as  $l_1 \rightarrow \infty$ ,

$$\sup_{f_+ \in \mathcal{F}_+} \sup_{\phi_+ \in \mathcal{P}_+} \left| \widehat{\mathcal{S}}_{+,l_1}(f_+, \phi_+) - \mathcal{S}_+(f_+, \phi_+) \right| \xrightarrow{\text{a.s.}} 0, \quad (13)$$

whereas for the negative class, as  $l_2 \rightarrow \infty$ ,

$$\sup_{f_- \in \mathcal{F}_-} \sup_{\phi_- \in \mathcal{P}_-} \left| \widehat{\mathcal{S}}_{-,l_2}(f_-, \phi_-) - \mathcal{S}_-(f_-, \phi_-) \right| \xrightarrow{\text{a.s.}} 0. \quad (14)$$

*Proof.* We first establish the weak convergence of the positive class-conditional statistical invariants. Since  $y_{+,i} = 1$  for all  $i = 1, \dots, l_1$ , Eqs. (9) and (10) give, for any  $f_+ \in \mathcal{F}_+$  and  $\phi_+ \in \mathcal{P}_+$ ,  $\widehat{\mathcal{S}}_{+,l_1}(f_+, \phi_+) - \mathcal{S}_+(f_+, \phi_+) = \langle f_+, \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i})K(x_{+,i}, \cdot) - \int_{\mathcal{X}} \phi_+(x)K(x, \cdot) dP_+(x) \rangle_{\mathcal{H}} - [\frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i}) - \int_{\mathcal{X}} \phi_+(x) dP_+(x)]$ , where the first equality follows from the reproducing property  $f_+(x) = \langle f_+, K(x, \cdot) \rangle_{\mathcal{H}}$ . Hence, by the Cauchy-Schwarz inequality,

$$\begin{aligned} & \left| \widehat{\mathcal{S}}_{+,l_1}(f_+, \phi_+) - \mathcal{S}_+(f_+, \phi_+) \right| \\ & \leq \|f_+\|_{\mathcal{H}} \left\| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i})K(x_{+,i}, \cdot) - \int_{\mathcal{X}} \phi_+(x)K(x, \cdot) dP_+(x) \right\|_{\mathcal{H}} \\ & \quad + \left| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i}) - \int_{\mathcal{X}} \phi_+(x) dP_+(x) \right|. \end{aligned} \quad (\text{C.1})$$

Since  $\|K(x, \cdot)\|_{\mathcal{H}} = \sqrt{K(x, x)} \leq \kappa$  and the predicates in  $\mathcal{P}_+$  are uniformly bounded, both  $\phi_+(x)K(x, \cdot)$  and  $\phi_+(x)$  are integrable. For each fixed  $\phi_+ \in \mathcal{P}_+$ , the samples  $x_{+,1}, \dots, x_{+,l_1}$  are i.i.d. from  $P_+$ . Therefore, the Hilbert-space and scalar strong laws of large numbers imply, respectively,  $\left\| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i})K(x_{+,i}, \cdot) - \int_{\mathcal{X}} \phi_+(x)K(x, \cdot) dP_+(x) \right\|_{\mathcal{H}} \xrightarrow{\text{a.s.}} 0$  and  $\left| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i}) - \int_{\mathcal{X}} \phi_+(x) dP_+(x) \right| \xrightarrow{\text{a.s.}} 0$ . Taking the supremum over  $\mathcal{F}_+$  and  $\mathcal{P}_+$  in Eq. (C.1) yields  $\sup_{f_+ \in \mathcal{F}_+} \sup_{\phi_+ \in \mathcal{P}_+} \left| \widehat{\mathcal{S}}_{+,l_1}(f_+, \phi_+) - \mathcal{S}_+(f_+, \phi_+) \right| \leq (\sup_{f_+ \in \mathcal{F}_+} \|f_+\|_{\mathcal{H}}) \max_{\phi_+ \in \mathcal{P}_+} \left\| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i})K(x_{+,i}, \cdot) - \int_{\mathcal{X}} \phi_+(x)K(x, \cdot) dP_+(x) \right\|_{\mathcal{H}} + \max_{\phi_+ \in \mathcal{P}_+} \left| \frac{1}{l_1} \sum_{i=1}^{l_1} \phi_+(x_{+,i}) - \int_{\mathcal{X}} \phi_+(x) dP_+(x) \right|$ . By assumption,  $\sup_{f_+ \in \mathcal{F}_+} \|f_+\|_{\mathcal{H}} < \infty$ . Moreover, since  $\mathcal{P}_+$  is finite, the two maxima converge to zero almost surely on a common probability-one event. Therefore, the empirical positive class-conditional statistical invariants converge uniformly to their population counterparts almost surely, establishing Eq. (13).

The weak convergence of the negative class-conditional statistical invariants follows analogously. Since  $y_{-,i} = 1$ , the samples  $x_{-,1}, \dots, x_{-,l_2}$  are i.i.d. from  $P_-$ , and  $\mathcal{F}_-$  and  $\mathcal{P}_-$  satisfy the same boundedness and finiteness conditions, the same argument yields Eq. (14).  $\square$

## Appendix D Proof of Proposition 3.1

### Proposition 3.1 (statement).

Consider the TLUSI problems in Eqs. (15) and (16). Under appropriate choices of  $\mathcal{P}_+$  and  $\mathcal{P}_-$ , TLUSI exhibits the following structural relationships with representative twin SVM models:

- (i) If  $\mathcal{P}_+ = \mathbf{I}_+$  and  $\mathcal{P}_- = \mathbf{I}_-$ , TLUSI can be transformed into the corresponding TBSVM formulation [40].
- (ii) If  $\mathcal{P}_+ = \mathbf{D}_+$  and  $\mathcal{P}_- = \mathbf{D}_-$ , where  $\mathbf{D}_+$  and  $\mathbf{D}_-$  are the within-class local weighting matrices, and the inter-class neighborhood matrices satisfy  $\mathbf{F}_+ = \mathbf{I}_+$  and  $\mathbf{F}_- = \mathbf{I}_-$ , TLUSI has the same principal optimization structure as the corresponding WTSVM formulation [41].
- (iii) Let  $\Sigma_+, \Sigma_- \in \mathbb{R}^{l \times l}$  be positive-semidefinite within-class structural matrices, and define  $\widetilde{\Sigma}_{\pm} = \text{diag}(\Sigma_{\pm}, 0)$ . Suppose that  $(\mathbf{H}_+^{\dagger} \mathbf{H}_+) \widetilde{\Sigma}_+ = \widetilde{\Sigma}_+$  and  $(\mathbf{H}_-^{\dagger} \mathbf{H}_-) \widetilde{\Sigma}_- = \widetilde{\Sigma}_-$ , where  $\mathbf{H}_{\pm}^{\dagger}$  denote the Moore-Penrose inverses. If  $\mathcal{P}_+ = \mathbf{I}_+ + \lambda_+ (\mathbf{H}_+^{\dagger})^{\top} \widetilde{\Sigma}_+ \mathbf{H}_+$  and  $\mathcal{P}_- = \mathbf{I}_- + \lambda_- (\mathbf{H}_-^{\dagger})^{\top} \widetilde{\Sigma}_- \mathbf{H}_-$ , with  $\lambda_+, \lambda_- \geq 0$ , the TLUSI invariant terms decompose into within-class squared fitting and structural regularization terms, reducing TLUSI to the corresponding STWSVM formulation [42].

*Proof.* Consider the TLUSI problems in Eqs. (15) and (16). It suffices to analyze the positive problem, since the negative result follows by interchanging  $(\mathbf{X}_+, l_1, c_1, c_3)$  and  $(\mathbf{X}_-, l_2, c_2, c_4)$ . We observe that  $\mathbf{H}_+ \mathbf{u}_1 = K(\mathbf{X}_+, \mathbf{X}) \mathbf{w}_1 + b_1 \mathbf{e}_1$ , and  $\mathbf{y}_+ = \mathbf{e}_1$ , where  $\mathbf{w}_1 \in \mathbb{R}^l$  is the kernel expansion coefficient vector. By introducing the translated bias  $\widetilde{b}_1 = b_1 - 1$  and defining  $\widetilde{\mathbf{u}}_1 = [\mathbf{w}_1^{\top}, \widetilde{b}_1]^{\top}$ , the invariant term can be written as  $(\mathbf{H}_+ \mathbf{u}_1 - \mathbf{y}_+)^{\top} \mathcal{P}_+ (\mathbf{H}_+ \mathbf{u}_1 - \mathbf{y}_+) = \widetilde{\mathbf{u}}_1^{\top} (\mathbf{H}_+^{\top} \mathcal{P}_+ \mathbf{H}_+) \widetilde{\mathbf{u}}_1$ .

and the constraint becomes  $\mathbf{H}_- \bar{\mathbf{u}}_1 - \bar{\xi}_2 \leq -\frac{1}{2} \mathbf{e}_2$ . Introduce the invertible transformation  $\bar{\mathbf{w}}_1 = 2\mathbf{w}_1$ ,  $\bar{b}_1 = 2\tilde{b}_1$ ,  $\bar{\xi}_2 = 2\xi_2$ ,  $\bar{\mathbf{u}}_1 = [\bar{\mathbf{w}}_1^\top, \bar{b}_1]^\top$ . Multiplying the transformed objective by four and removing the variable-independent constant yields the equivalent problem

$$\begin{aligned} \min_{\bar{\mathbf{u}}_1, \bar{\xi}_2} \quad & \frac{c_3}{2} \bar{\mathbf{u}}_1^\top \mathbf{H}_+^\top \mathcal{P}_+ \mathbf{H}_+ \bar{\mathbf{u}}_1 + \frac{1}{2} \|\bar{\mathbf{u}}_1\|_2^2 + 2\bar{b}_1 + 2c_1 \mathbf{e}_2^\top \bar{\xi}_2 \\ \text{s.t.} \quad & -\mathbf{H}_- \bar{\mathbf{u}}_1 + \bar{\xi}_2 \geq \mathbf{e}_2, \quad \bar{\xi}_2 \geq \mathbf{0}. \end{aligned} \quad (\text{D.1})$$

Thus, the three relationships in Proposition 3.1 differ only in the specification of  $\mathcal{P}_+$ .

(i) *Relationship with TBSVM..* When  $\mathcal{P}_+ = \mathbf{I}_+$ , the quadratic invariant term in Eq. (D.1) reduces to  $\bar{\mathbf{u}}_1^\top \mathbf{H}_+^\top \mathbf{H}_+ \bar{\mathbf{u}}_1 = \|\mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2 = \|K(\mathbf{X}_+, \mathbf{X}) \bar{\mathbf{w}}_1 + \bar{b}_1 \mathbf{e}_1\|_2^2$ . Setting  $C_1 = 2c_1$ , Eq. (D.1) becomes

$$\begin{aligned} \min_{\bar{\mathbf{u}}_1, \bar{\xi}_2} \quad & \frac{c_3}{2} \|\mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2 + \frac{1}{2} \|\bar{\mathbf{u}}_1\|_2^2 + 2\bar{b}_1 + C_1 \mathbf{e}_2^\top \bar{\xi}_2 \\ \text{s.t.} \quad & -\mathbf{H}_- \bar{\mathbf{u}}_1 + \bar{\xi}_2 \geq \mathbf{e}_2, \quad \bar{\xi}_2 \geq \mathbf{0}. \end{aligned} \quad (\text{D.2})$$

Except for the linear bias correction term  $2\bar{b}_1$  induced by the probability calibration ( $b_1 = \tilde{b}_1 + 1$ ), Eq. (D.2) is mathematically identical to the primal TBSVM formulation. Hence, TLUSI and TBSVM have the same principal optimization structure.

(ii) *Relationship with WTSVM..* When  $\mathcal{P}_+ = \mathbf{D}_+ = \text{diag}(d_1^+, \dots, d_{l_1}^+)$ , with  $d_i^+ \geq 0$ , and the cross-class neighborhood matrix satisfies  $\mathbf{F}_- = \mathbf{I}_-$ , the quadratic invariant term in Eq. (D.1) expands to  $\bar{\mathbf{u}}_1^\top \mathbf{H}_+^\top \mathbf{D}_+ \mathbf{H}_+ \bar{\mathbf{u}}_1 = \sum_{i=1}^{l_1} d_i^+ (K(\mathbf{x}_{+,i}, \mathbf{X}) \bar{\mathbf{w}}_1 + \bar{b}_1)^2 = \|\mathbf{D}_+^{1/2} \mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2$ . Consequently, Eq. (D.1) becomes

$$\begin{aligned} \min_{\bar{\mathbf{u}}_1, \bar{\xi}_2} \quad & \frac{c_3}{2} \|\mathbf{D}_+^{1/2} \mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2 + \frac{1}{2} \|\bar{\mathbf{u}}_1\|_2^2 + 2\bar{b}_1 + C_1 \mathbf{e}_2^\top \bar{\xi}_2 \\ \text{s.t.} \quad & -\mathbf{F}_- \mathbf{H}_- \bar{\mathbf{u}}_1 + \bar{\xi}_2 \geq \mathbf{e}_2, \quad \bar{\xi}_2 \geq \mathbf{0}. \end{aligned} \quad (\text{D.3})$$

With  $C_1 = 2c_1$ , Eq (D.3) therefore has the same structure as the corresponding WTSVM problem up to the linear bias correction  $2\bar{b}_1$ .

(iii) *Relationship with STWSVM..* When  $\mathcal{P}_+ = \mathbf{I}_+ + \lambda_+ (\mathbf{H}_+^\dagger)^\top \tilde{\Sigma}_+ \mathbf{H}_+^\dagger$ , the matrix  $\mathbf{H}_+^\dagger \mathbf{H}_+$  is the orthogonal projector onto the row space of  $\mathbf{H}_+$ . The assumed compatibility condition, together with the symmetry of  $\tilde{\Sigma}_+$ , implies  $(\mathbf{H}_+^\dagger \mathbf{H}_+)^\top \tilde{\Sigma}_+ (\mathbf{H}_+^\dagger \mathbf{H}_+) = \tilde{\Sigma}_+$ . Therefore,  $\mathbf{H}_+^\top \mathcal{P}_+ \mathbf{H}_+ = \mathbf{H}_+^\top \mathbf{H}_+ + \lambda_+ (\mathbf{H}_+^\dagger \mathbf{H}_+)^\top \tilde{\Sigma}_+ (\mathbf{H}_+^\dagger \mathbf{H}_+) = \mathbf{H}_+^\top \mathbf{H}_+ + \lambda_+ \tilde{\Sigma}_+$ . Since  $\tilde{\Sigma}_+ = \text{diag}(\Sigma_+, 0)$ , the invariant term decomposes as

$$\bar{\mathbf{u}}_1^\top \mathbf{H}_+^\top \mathcal{P}_+ \mathbf{H}_+ \bar{\mathbf{u}}_1 = \|\mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2 + \lambda_+ \bar{\mathbf{w}}_1^\top \Sigma_+ \bar{\mathbf{w}}_1. \quad (\text{D.4})$$

Substituting Eq. (D.4) into Eq. (D.1) and dividing the objective by  $c_3 > 0$  yields  $\min_{\bar{\mathbf{u}}_1, \bar{\xi}_2} \frac{1}{2} \|\mathbf{H}_+ \bar{\mathbf{u}}_1\|_2^2 + \frac{\lambda_+}{2} \bar{\mathbf{w}}_1^\top \Sigma_+ \bar{\mathbf{w}}_1 + \frac{1}{2c_3} \|\bar{\mathbf{u}}_1\|_2^2 + \frac{2c_1}{c_3} \mathbf{e}_2^\top \bar{\xi}_2 + \frac{2}{c_3} \bar{b}_1$  s.t.  $-\mathbf{H}_- \bar{\mathbf{u}}_1 + \bar{\xi}_2 \geq \mathbf{e}_2$ ,  $\bar{\xi}_2 \geq \mathbf{0}$ . By matching parameters  $\rho_1 = 2c_1/c_3$ ,  $\rho_2 = 1/c_3$ , and  $\rho_3 = \lambda_+$ , this matches the primal formulation of STWSVM, subject to the linear bias correction  $(2/c_3)\bar{b}_1$ .

The same arguments apply symmetrically to the negative TLUSI problem with  $\mathcal{P}_-$ ,  $\mathbf{D}_-$ ,  $\tilde{\Sigma}_-$ , and the corresponding negative parameters. This completes the proof of Proposition 3.1.  $\square$

## Appendix E Proof of Theorem 3.2

### Theorem 3.2 (statement).

Let  $\Psi(\mathbf{z}) = \frac{1}{2} \mathbf{z}^\top \mathbf{Q} \mathbf{z} - \mathbf{g}^\top \mathbf{z}$ ,  $\mathcal{Z} = [0, C]^n$ , where  $\mathbf{Q} \geq \mathbf{0}$  and  $C > 0$ . Suppose that  $0 < \omega < 2$  and that  $\{\mathbf{z}^{(k)}\}$  is the sequence generated by Algorithm 1 without premature termination. Then:

- (i)  $\mathbf{z}^{(k)} \in \mathcal{Z}$  for every  $k$ , and  $\{\Psi(\mathbf{z}^{(k)})\}$  converges monotonically to the optimal objective value;
- (ii)  $\lim_{k \rightarrow \infty} \|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\|_2 = 0$ ,  $\lim_{k \rightarrow \infty} s_k = 0$ ;
- (iii) every accumulation point of  $\{\mathbf{z}^{(k)}\}$  solves Eq. (21), and  $\lim_{k \rightarrow \infty} \text{dist}(\mathbf{z}^{(k)}, \mathcal{Z}^*) = 0$ ,  $\mathcal{Z}^* = \arg \min_{\mathbf{z} \in \mathcal{Z}} \Psi(\mathbf{z})$ .

If  $\mathbf{Q} > \mathbf{0}$ , the optimal solution is unique and the entire sequence converges to that solution.

*Proof.* (i) Since  $\mathbf{z}^{(0)} = \mathbf{0} \in \mathcal{Z}$ , let  $\mathbf{z}^{(k,i-1)}$  denote the intermediate vector before coordinate  $i$  is updated during sweep  $k$ , with  $\mathbf{z}^{(k,0)} = \mathbf{z}^{(k)}$  and  $\mathbf{z}^{(k,n)} = \mathbf{z}^{(k+1)}$ . From Algorithm 1,  $r_i = \sum_{j < i} Q_{ij} z_j^{(k+1)} + \sum_{j \geq i} Q_{ij} z_j^{(k)} - g_i = \nabla_i \Psi(\mathbf{z}^{(k,i-1)})$ . Hence the coordinate update is equivalently

$$z_i^{(k,i)} = \Pi_{[0,C]} \left( z_i^{(k,i-1)} - \frac{\omega}{Q_{ii} + \rho} \nabla_i \Psi(\mathbf{z}^{(k,i-1)}) \right), \quad (\text{E.1})$$

while all other coordinates remain unchanged. Since  $\mathbf{Q} \geq \mathbf{0}$  gives  $Q_{ii} \geq 0$  and  $\rho \geq \epsilon > 0$  by construction, the denominator is strictly positive. The projection therefore guarantees  $\mathbf{z}^{(k)} \in \mathcal{Z}$  for every  $k$ . Let  $\Delta_i^{(k)} = z_i^{(k,i)} - z_i^{(k,i-1)}$ . The variational inequality for the projection in Eq. (E.1), evaluated at  $z_i^{(k,i-1)} \in [0, C]$ , gives

$$\nabla_i \Psi(\mathbf{z}^{(k,i-1)}) \Delta_i^{(k)} + \frac{Q_{ii} + \rho}{\omega} |\Delta_i^{(k)}|^2 \leq 0. \quad (\text{E.2})$$

Since  $\Psi$  is quadratic and only coordinate  $i$  changes,

$$\Psi(\mathbf{z}^{(k,i)}) = \Psi(\mathbf{z}^{(k,i-1)}) + \nabla_i \Psi(\mathbf{z}^{(k,i-1)}) \Delta_i^{(k)} + \frac{Q_{ii}}{2} |\Delta_i^{(k)}|^2. \quad (\text{E.3})$$

Combining Eqs. (E.2) and (E.3) yields

$$\Psi(\mathbf{z}^{(k,i-1)}) - \Psi(\mathbf{z}^{(k,i)}) \geq \left( \frac{Q_{ii} + \rho}{\omega} - \frac{Q_{ii}}{2} \right) |\Delta_i^{(k)}|^2. \quad (\text{E.4})$$

Because  $0 < \omega < 2$ , we have  $\frac{Q_{ii} + \rho}{\omega} - \frac{Q_{ii}}{2} = Q_{ii} \left( \frac{1}{\omega} - \frac{1}{2} \right) + \frac{\rho}{\omega} > 0$ . Since the number of coordinates is finite, there exists  $\eta > 0$  such that, after summing Eq. (E.4) over one complete sweep,

$$\Psi(\mathbf{z}^{(k)}) - \Psi(\mathbf{z}^{(k+1)}) \geq \eta \sum_{i=1}^n |\Delta_i^{(k)}|^2 = \eta \|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\|_2^2. \quad (\text{E.5})$$

Thus  $\{\Psi(\mathbf{z}^{(k)})\}$  is nonincreasing. Since  $\mathcal{Z}$  is compact and  $\Psi$  is continuous, the objective is bounded below on  $\mathcal{Z}$  and therefore converges to a finite limit. We show in part (iii) that this limit is the optimal objective value.

(ii) Summing Eq. (E.5) over  $k$  gives  $\sum_{k=0}^{\infty} \|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\|_2^2 < \infty$ . Consequently,

$$\|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\|_2 \rightarrow 0. \quad (\text{E.6})$$

In particular, since  $d_k = \frac{\|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\|_2}{1 + \|\mathbf{z}^{(k)}\|_2}$ , we also have  $d_k \rightarrow 0$ . The convergence  $s_k \rightarrow 0$  follows below from the optimality of all accumulation points.

(iii) Since  $\{\mathbf{z}^{(k)}\} \subset \mathcal{Z}$  and  $\mathcal{Z}$  is compact, the sequence has an accumulation point. Let  $\mathbf{z}^{(k_j)} \rightarrow \bar{\mathbf{z}}$ . By Eq. (E.6) and Eq. (E.5), each coordinate change  $\Delta_i^{(k_j)}$  tends to zero. Hence every intermediate vector  $\mathbf{z}^{(k_j,i)}$  in the same sweep also converges to  $\bar{\mathbf{z}}$ . Passing to the limit in Eq. (E.1) gives, for each  $i = 1, \dots, n$ ,  $\bar{z}_i = \Pi_{[0,C]}(\bar{z}_i - \frac{\omega}{Q_{ii} + \rho} \nabla_i \Psi(\bar{\mathbf{z}}))$ . Since

$$\omega/(Q_{ii} + \rho) > 0, \text{ this fixed-point relation is equivalent to } \begin{cases} \nabla_i \Psi(\bar{\mathbf{z}}) \geq 0, & \bar{z}_i = 0, \\ \nabla_i \Psi(\bar{\mathbf{z}}) = 0, & 0 < \bar{z}_i < C, \\ \nabla_i \Psi(\bar{\mathbf{z}}) \leq 0, & \bar{z}_i = C. \end{cases} \text{ These are exactly the first-order}$$

optimality conditions for minimizing the convex function  $\Psi$  over  $\mathcal{Z}$ . Therefore,  $\bar{\mathbf{z}} \in \mathcal{Z}^*$ . Since the same argument applies to every convergent subsequence, every accumulation point of  $\{\mathbf{z}^{(k)}\}$  is globally optimal.

Because  $\Psi(\mathbf{z}^{(k)})$  has a limit and  $\mathbf{z}^{(k_j)} \rightarrow \bar{\mathbf{z}} \in \mathcal{Z}^*$ , continuity gives  $\Psi(\mathbf{z}^{(k_j)}) \rightarrow \Psi(\bar{\mathbf{z}}) = \min_{\mathbf{z} \in \mathcal{Z}} \Psi(\mathbf{z})$ . Hence the entire objective sequence converges to the optimal objective value, which completes part (i).

We next show convergence to the solution set. If  $\text{dist}(\mathbf{z}^{(k)}, \mathcal{Z}^*) \not\rightarrow 0$ , then there exist  $\delta > 0$  and a subsequence satisfying  $\text{dist}(\mathbf{z}^{(k_j)}, \mathcal{Z}^*) \geq \delta$ . By compactness, this subsequence has a further convergent subsequence. Its limit is an accumulation point of  $\{\mathbf{z}^{(k)}\}$  and hence belongs to  $\mathcal{Z}^*$ , which contradicts the inequality above. Therefore,

$$\text{dist}(\mathbf{z}^{(k)}, \mathcal{Z}^*) \rightarrow 0. \quad (\text{E.7})$$

For every  $\mathbf{z} \in \mathcal{Z}^*$ , the first-order optimality condition is equivalently  $\mathbf{z} = \Pi_{\mathcal{Z}}(\mathbf{z} - \mathbf{Q}\mathbf{z} + \mathbf{g})$ . The projected-gradient residual used in Algorithm 1 is continuous. Since Eq. (E.7) holds and this residual vanishes on  $\mathcal{Z}^*$ ,  $s_k = \|\mathbf{z}^{(k+1)} - \Pi_{\mathcal{Z}}(\mathbf{z}^{(k+1)} - \mathbf{Q}\mathbf{z}^{(k+1)} + \mathbf{g})\|_{\infty} \rightarrow 0$ . This completes part (ii).

Finally, if  $\mathbf{Q} > \mathbf{0}$ , then  $\Psi$  is strictly convex on  $\mathcal{Z}$ , so  $\mathcal{Z}^*$  contains a unique minimizer. Since every accumulation point of  $\{\mathbf{z}^{(k)}\}$  equals this minimizer, the entire sequence converges to the unique optimal solution.  $\square$

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